

The pyramid — height, apothem and surface area

Three lengths, one right triangle, and every pyramid question follows.

LESSON 33–34

2 × 40 MIN

GEOMETRY 11, §15

IGCSE E13.X

LESSON CLOCK

14'

24'

24'

20'

16'

4

The elements	14 min
The two right triangles	24 min
Surface area	24 min
Working backwards	20 min
Practice	16 min
Homework	4 min

LEARNING OBJECTIVES

- Name the elements of a pyramid and of a regular pyramid.
- Relate the height, apothem, slant height and lateral edge.
- Compute the lateral and total surface area.
- Solve for a missing element from a given area.

TEXTBOOK

National	pp. 161–172
Cambridge	Core & Extended, pp. 323–330
Workbook	Exercise 15.1

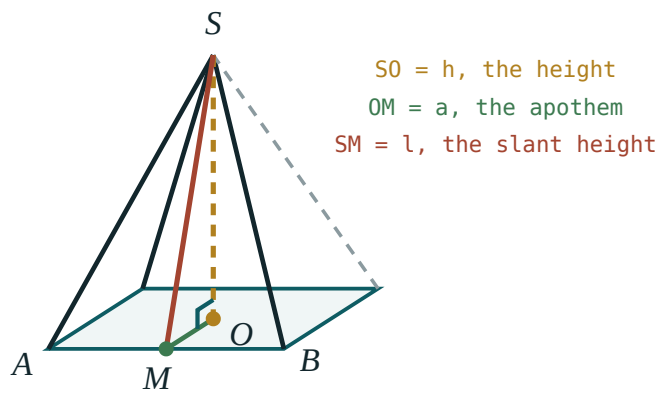
01

Explanation

The elements

REGULAR PYRAMID

Its base is a **regular polygon** and its apex lies directly above the base **centre**. Then all the lateral edges are equal, and all the lateral faces are congruent isosceles triangles.



SO = h, the height
OM = a, the apothem
SM = l, the slant height

$l^2 = h^2 + a^2 \cdot S_{lat} = \frac{1}{2}Pl$

The height to the centre, the apothem across the base,
the slant height up the face.

Why the lateral edges are equal: their projections are all radii of the circumcircle of the base, hence equal — and equal projections give equal obliques, by Grade 10 Quarter III.

The two right triangles

Every pyramid calculation lives in one of two right triangles, both with the height *h* as a leg:

$l^2 = h^2 + a^2$ (slant height, apothem) $e^2 = h^2 + R^2$ (lateral edge, circumradius)

SQUARE BASE, SIDE <i>s</i>	VALUE
apothem <i>a</i>	$\frac{s}{2}$
circumradius <i>R</i>	$\frac{s\sqrt{2}}{2}$
slant height <i>l</i>	$\sqrt{h^2 + (s/2)^2}$
lateral edge <i>e</i>	$\sqrt{h^2 + s^2/2}$

THE APOTHEM IS NOT THE CIRCUMRADIUS

For a square of side 10: apothem 5, circumradius $5\sqrt{2} \approx 7.07$. Using the wrong one is the standard error, and it makes the slant height and the lateral edge swap places.

Surface area

$$S_{lat} = \frac{1}{2} P \cdot l \quad S_{total} = B + \frac{1}{2} Pl$$

Each lateral face is a triangle of base one side and height l ; there are as many as the base has sides, so the total is $\frac{1}{2}$ of the perimeter times l .

Example. Square base 10, height 12: $a = 5$, $l = 13$, $P = 40$, $B = 100$. So $S_{lat} = 260$ and $S = 360$.

ONLY FOR A REGULAR PYRAMID

$\frac{1}{2} Pl$ assumes every lateral face has the same slant height. For an irregular pyramid the faces must be computed one at a time.

Working backwards

Given the area, recover a length by reversing the formula:

$$l = \frac{2S_{lat}}{P} \quad h = \sqrt{l^2 - a^2}$$

Example. A square pyramid with base 12 has lateral area 240. Then $l = \frac{480}{48} = 10$,

$a = 6$, so $h = \sqrt{100 - 36} = 8$.

02 Worked examples

EXAMPLE 1 A regular square pyramid has base edge 10 and height 12. Find the slant height, the lateral edge and the total surface area.

$$① \quad a = 5, \text{ so } l = \sqrt{144 + 25} = 13.$$

$$② \quad R = 5\sqrt{2}, \text{ so } e = \sqrt{144 + 50} = \sqrt{194} \approx 13.9.$$

$$③ \quad S_{lat} = \frac{1}{2} \times 40 \times 13 = 260$$

$$④ \quad S = 100 + 260 = 360$$

ANSWER

$$l = 13, e \approx 13.9, S = 360$$

EXAMPLE 2 A regular hexagonal pyramid has base edge 6 and slant height 10. Find the lateral surface area and the height.

$$① \quad P = 36, \text{ so } S_{lat} = \frac{1}{2} \times 36 \times 10 = 180.$$

$$② \quad \text{Apothem of a regular hexagon of side 6: } 3\sqrt{3} \approx 5.196.$$

$$③ \quad h = \sqrt{100 - 27} = \sqrt{73} \approx 8.54$$

ANSWER

$$S_{lat} = 180; h \approx 8.54$$

EXAMPLE 3 A square pyramid with base 12 has lateral surface area 240. Find its height.

$$① \quad l = \frac{2 \times 240}{48} = 10$$

$$② \quad a = 6$$

$$③ \quad h = \sqrt{100 - 36} = 8$$

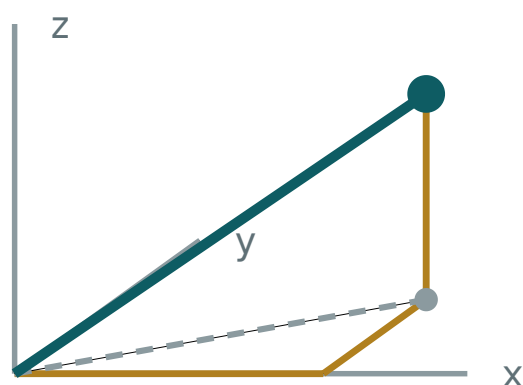
ANSWER

8

03 Interactive model

Build the pyramid net from Quarter I and measure the slant height directly.

Inside a pyramid



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Pyramid	Piramida	Пирамида
Apex	Uchi	Вершина
Regular pyramid	Muntazam piramida	Правильная пирамида
Height of a pyramid	Piramida balandligi	Высота пирамиды
Apothem of the base	Asos apofemasi	Апофема основания
Slant height	Yon apofema	Апофема
Lateral edge	Yon qirra	Боковое ребро
Lateral surface area	Yon sirt yuzasi	Площадь боковой поверхности
Base perimeter	Asos perimetri	Периметр основания

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

In a regular pyramid the apex is above:

l^2 equals:

S_{lat} of a regular pyramid:

For a square base of side 10 the apothem is:

And the circumradius is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Square base 10, height 12: the slant height

ANSWER 13

2. Square base 8, height 3: the slant height

ANSWER 5

3. Square base 6, apothem

ANSWER 3

4. Square base 6, circumradius

ANSWER $3\sqrt{2}$

5. S_{lat} with $P = 24$, $l = 10$

ANSWER 120

6. S_{total} with $B = 36$, $P = 24$, $l = 10$

ANSWER 156

7. Square base 10, height 12: the lateral area

ANSWER 260

Medium · the standard question

7 PROBLEMS

1. Square base 10, height 12: total area

ANSWER 360

2. Square base 10, height 12: the lateral edge

ANSWER $\sqrt{194} \approx 13.9$

3. Hexagonal base 6, slant height 10: lateral area

ANSWER 180

4. Same: the height

ANSWER $\sqrt{73} \approx 8.54$

5. Square base 12, lateral area 240: the height

ANSWER 8

6. Square base 16, height 6: total area

ANSWER 576

7. Square base 14, lateral edge 25: the height

ANSWER 24

Hard · stretch and reason

7 PROBLEMS

1. A regular triangular pyramid, base edge 6, height 4: the total area

ANSWER $9\sqrt{3} + 3\sqrt{57} \approx 38.2$

2. Square base s , height h : prove $e^2 = h^2 + \frac{s^2}{2}$

ANSWER $R = \frac{s\sqrt{2}}{2}$

3. A square pyramid of total area 384 with base 12: the height

ANSWER 8

4. A regular hexagonal pyramid, edge 4, height 6: the volume

ANSWER $48\sqrt{3} \approx 83.1$

5. A pyramid's lateral faces make 60° with the base, base edge 10: the height

ANSWER $5\sqrt{3}$

6. Show that $e > l > h$ in every regular pyramid

ANSWER Each is the hypotenuse of the previous

7. A square pyramid with all edges equal to a : its height

ANSWER $\frac{a}{\sqrt{2}} \approx 0.707a$

07 Homework

Homework — 5 tasks

Say which triangle you are using — apothem or circumradius — on every line.

1. A regular square pyramid has base edge 16 and height 15. Find the slant height, the lateral edge and the total surface area.
2. A regular hexagonal pyramid has base edge 5 and slant height 12. Find the lateral area and the height.

3. A square pyramid with base 20 has lateral area 520 . Find the slant height and the height.
4. A regular triangular pyramid has base edge 8 and height 6 . Find the slant height and the total surface area.
5. Prove that in a regular pyramid the lateral edges are equal.